

Applied Bayesian Statistics in Animal & Biological Sciences

Henk J. van Lingen

hvanling@uoguelph.ca

May 11 - May 15, 2026



Course Sessions in Applied Bayesian Statistics

- ① Mon May 11, 11:00-12:00 - Introduction to Bayesian reasoning in Animal Biosciences
- ② Mon May 11, 1:00-3:00 - (Hierarchical) linear prediction models
- ③ Tue May 12, 1:00-3:00 - Linear models & multiple comparisons
- ④ Wed May 13, 1:00-3:00 - Non-linear prediction and heteroscedasticity
- ⑤ Thu May 14, 1:00-3:00 - Discrete data modeling
- ⑥ Fri May 15, 1:00-3:00 - Parameter calibration of ordinary differential equation models

Miscellaneous Course Logistics and Information

- What to expect: link to FAQ
- Every afternoon session includes a lecture and exercises
- Outside of course hours you can find me in ANNU 021
- Practicing model coding and simulation outside of class will help you get the most out of this course!
- You are encouraged to develop models using your own data!
- If you never used R before, reach out to obtain a basic tutorial
- Drinks & Bites on Friday, May 15, from 3:00 pm

Installing R and RSTAN

Make sure you have RSTAN up and running before the course starts:

- Update to or install a recent version of R:
`https://cran.r-project.org/`
- Install Rtools (for windows users only) compatible with your version of R: `https://cran.r-project.org/bin/windows/Rtools/`
- Install the RSTAN package by running: `install.packages("rstan")`
- Install the loo package by running: `install.packages("loo")`
- Run the "Verify_RSTAN_works.R" code I shared to verify everything works